

BLOEMFONTEIN INTL, South AfricaWMO: **684420**Lat: **29.1039S**Lon: **26.2981E**Elev: **4439**StdP: **12.49**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	26.0	5.0	8.4	61.5	9.2	10.4	56.9	21.6	57.1	18.8	57.1	1.5	210	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.0	93.4	59.8	91.2	59.7	89.2	59.7	67.4	79.3	66.4	78.6	65.3	77.8	10.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	106.9	70.9	62.8	101.0	70.0	61.5	96.3	69.4	34.7	78.3	33.7	78.5	32.8	77.6	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	17.4	15.3	DB	18.9	97.1	2.7	3.0	16.9	99.3	15.4	101.1	13.8	102.8	11.9	105.0	(4)
(5)			WB	17.3	70.1	2.7	2.1	15.4	71.7	13.8	72.9	12.2	74.1	10.2	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	72.8	71.5	67.7	60.0	53.0	46.8	46.2	51.3	58.8	64.6	67.9	71.5	(6)	
	DBStd	10.65	4.38	4.02	4.52	4.91	5.00	4.45	4.68	6.07	6.24	5.75	5.88	4.60	(7)	
	HDD50	342	0	0	0	2	29	114	135	56	6	0	0	0	(8)	
	HDD65	2445	4	5	25	161	373	545	584	425	202	78	38	5	(9)	
	CDD50	4340	706	603	548	301	121	19	17	98	270	453	537	667	(10)	
	CDD65	966	245	188	108	10	0	0	0	1	16	66	125	207	(11)	
	CDH74	12995	2682	1956	1360	338	56	1	2	102	695	1327	1885	2591	(12)	
	CDH80	4997	1180	777	428	48	2	0	0	8	191	467	764	1132	(13)	
(14) Wind	WSAvg	6.0	7.2	6.2	5.5	4.6	4.1	4.2	4.5	5.7	6.5	7.4	8.0	7.9	(14)	
(15) Precipitation	PrecAvg	21.6	3.4	3.7	3.0	1.9	0.8	0.5	0.3	0.5	0.7	1.8	2.5	2.6	(15)	
	PrecMax	44.0	9.0	20.9	7.9	5.7	3.5	3.7	1.9	4.6	7.4	8.5	7.5	6.4	(16)	
	PrecMin	11.1	0.4	0.2	0.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	(17)	
	PrecStd	7.2	2.3	3.3	1.5	1.4	0.9	0.7	0.5	0.9	1.2	1.7	1.8	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.7	93.5	91.5	85.6	78.6	71.9	73.2	80.8	89.3	91.7	93.6	97.0	(19)	
		MCWB	59.9	61.3	60.1	58.0	54.2	49.8	48.5	53.1	56.7	57.1	58.8	59.5	(20)	
	2%	DB	93.1	91.1	87.9	80.8	75.3	69.4	69.8	76.9	85.2	88.2	91.1	93.0	(21)	
		MCWB	60.0	61.0	60.6	57.7	52.8	48.9	47.9	51.1	55.0	56.6	58.5	59.7	(22)	
	5%	DB	89.9	88.1	84.6	78.0	72.9	66.4	67.1	73.6	81.9	85.7	88.0	89.6	(23)	
		MCWB	61.1	61.6	60.8	57.0	52.1	47.7	47.2	50.0	53.3	56.2	58.0	59.5	(24)	
	10%	DB	87.0	85.5	82.0	75.2	69.8	64.1	64.2	70.0	77.9	82.2	84.6	86.4	(25)	
		MCWB	61.6	62.2	61.0	56.6	51.7	46.8	46.3	48.8	52.4	55.7	57.6	59.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	69.2	67.2	63.7	58.3	53.3	52.3	56.3	60.4	63.7	65.2	68.1	(27)	
		MCDB	80.8	81.5	79.2	74.5	70.4	65.5	65.8	73.8	80.0	77.1	78.5	81.2	(28)	
	2%	WB	67.4	67.4	65.4	61.5	56.3	51.2	50.2	54.0	57.7	61.4	63.7	66.2	(29)	
		MCDB	78.7	79.3	77.9	72.9	68.6	64.6	65.3	70.9	77.3	76.3	77.4	79.6	(30)	
	5%	WB	66.4	66.3	64.1	60.0	54.5	49.5	48.4	52.2	55.9	59.8	62.4	64.6	(31)	
		MCDB	78.3	78.5	77.0	71.9	67.3	62.9	64.0	68.8	75.2	75.4	76.9	78.6	(32)	
	10%	WB	65.1	65.1	63.1	58.7	52.9	47.6	46.8	50.2	54.0	58.4	61.1	63.5	(33)	
		MCDB	77.4	77.1	76.0	70.6	65.8	61.0	62.4	66.3	72.8	74.3	75.6	77.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	26.0	25.3	26.6	28.6	31.9	33.6	34.8	34.9	34.8	31.1	29.3	28.1	(35)	
		MCDBR	31.9	31.5	32.1	33.5	36.9	37.4	38.7	39.1	40.4	36.5	34.0	33.6	(36)	
	5% WB	MCWBR	10.9	10.7	11.8	14.5	17.4	19.9	20.5	19.2	18.0	14.6	12.6	12.0	(37)	
		MCWBR	23.3	23.2	25.3	26.3	29.7	32.1	34.5	33.3	33.4	28.4	27.0	26.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.350	0.340	0.304	0.281	0.265	0.264	0.315	0.351	0.368	0.328	0.331	(40)	
		taud	2.434	2.422	2.422	2.504	2.546	2.556	2.546	2.384	2.276	2.268	2.432	2.463	(41)	
	Ebn at Noon		316	309	301	298	291	290	295	289	292	299	319	321	(42)	
		Edh at Noon	39	38	36	30	26	24	25	33	41	44	39	38	(43)	
(44) All-Sky Solar Radiation	RadAvg		2369	2130	1868	1515	1279	1142	1251	1512	1878	2173	2418	2498	(44)	
	RadStd		188	196	106	90	67	57	44	82	83	114	138	130	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.22	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+1.22	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page